Stimulating Static Routing protocol Configuration using CISCO packet Tracer

Exp.no: 11a

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Aim:

To stimulate and configure static Routing protocol using CISCO packet Tracer.

Static Routing:

Static Routing is a method of routing in which the routes are manually configured by the network administrator. The administrator specifies the destination network, subnet mask and next-hop address. In static routing, the router does not automatically learn routes from the routers. Therefore, the administrator must configure the required routes manually.

Static routers can also be configured with different administrative Distance. A route with lower administrative distance is preferred over a route with a higher administrative distance.

procedure:

(1) Create the required network topology in CISCO packet Tracer.

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(ii) Connect the devices using appropriate cables.

(iii) Configure the IP Addresses and subnet masks for all pc's and routers.

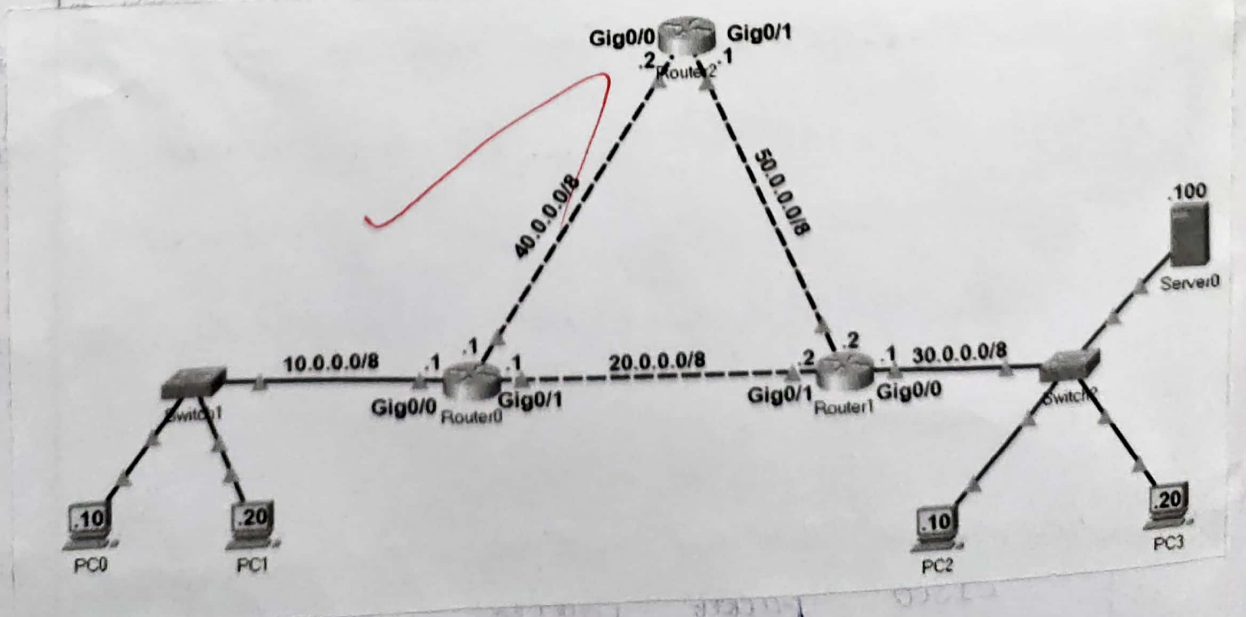
(iv) Enable router interfaces and configure static routes in each router.

(v) Configure a primary route with the lower Administrative Distance

(vi) Configure a backup route with a higher Administrative Distance.

(vii) Display the static routing table using,
show ip route static

(viii) Test the connectivity between pc's using ping. Then check path followed by packet using tracer.



Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configure commands, one per line.  End with CNTL/Z.
Router(config)#ip route 30.0.0.0 255.0.0.0 20.0.0.2 10 Primary route
Router(config)#ip route 30.0.0.0 255.0.0.0 40.0.0.2 20 Backup route
Router(config)#ip route 30.0.0.100 255.255.255.255 40.0.0.2 10 Primary route
Router(config)#ip route 30.0.0.100 255.255.255.255 20.0.0.2 20 Backup route
Router(config)#ip route 50.0.0.0 255.0.0.0 40.0.0.2 10 Primary route
Router(config)#ip route 50.0.0.0 255.0.0.0 20.0.0.2 20 Backup route
Router(config)#exit
Router#show ip route static
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
S       30.0.0.0/8 [10/0] via 20.0.0.2
S       30.0.0.100/32 [10/0] via 40.0.0.2
S    50.0.0.0/8 [10/0] via 40.0.0.2

Router#
```

Router adds only primary routes to the routing table.

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 10.0.0.0 255.0.0.0 20.0.0.1 10 main route
Router(config)#ip route 10.0.0.0 255.0.0.0 50.0.0.1 20 backup route
Router(config)#ip route 40.0.0.0 255.0.0.0 20.0.0.1 10 main route
Router(config)#ip route 40.0.0.0 255.0.0.0 50.0.0.1 20 backup route
Router(config)#exit
Router#show ip route static
S    10.0.0.0/8 [10/0] via 20.0.0.1
S    40.0.0.0/8 [10/0] via 20.0.0.1

Router#
```

Only main routes are added to the routing table.

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Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>enable
Router#show ip route static
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
S       30.0.0.0/8 [10/0] via 20.0.0.2
S       30.0.0.100/32 [10/0] via 40.0.0.2
S    50.0.0.0/8 [10/0] via 40.0.0.2

Router#
```

PC1

Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End wit
Router(config)#ip route 10.0.0.0 255.0.0.0 40.0.0.1
Router(config)#ip route 30.0.0.0 255.0.0.0 50.0.0.2
Router(config)#exit
Router#show ip route static
S    10.0.0.0/8 [1/0] via 40.0.0.1
S    30.0.0.0/8 [1/0] via 50.0.0.2
```

Result :

The static Routing protocol was stimulated and configured successfully using CISCO packet Tracer.

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